

EEG Classification Based on Variance

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Abstract —Brain computer interface (BCI) establishes a communication between human brain and an external assistive device via a computer. This communication system is mainly used for physically challenged individuals as well as it can be used to command robot to do task depending on mental thought. In this paper, we are working on the Electroencephalogram (EEG) signals based BCI system because the EEG signals having high temporal resolution and it is non-invasive. The objective of the paper is to enhance the classification accuracy for the movement imagery under the unsupervised learning. We have selected four preprocessing conventional filters namely Chebyshev filter, Butter worth filter, FIR bandpass filter and Elliptic filter to compare the performance in terms of classification accuracy. The filtered data is segmented and statistical parameter variance has been calculated in time domain. The difference of variance of the channels C3 and C4 has been taken as a feature. The Fisher linear discriminant analysis (FLDA) has been used to classify the feature matrix. The Elliptic filter achieves 84.3% classification accuracy for training data and 86.45% for testing data between the left and right hand movement imagination.

Index Terms— BCI (Brain Computer Interface), EEG (ElectroEncephalogram) signals, FIR (Finite Impulse Response), LC (Linear Classifier), FLDA (Fisher Linear Discriminant Analysis).

I. INTRODUCTION

BCI also called brain machine interface (BMI). It is a challenging problem that forms a wider area of research which interlinks thought to action. BCI will potentially provide a link between the brain and the physical world without any physical contact. For example, in the case of cursor control, the signal is transmitted directly from the brain to the mechanism directing the cursor, rather than taking the normal route through the body's neuromuscular system from the brain to the finger on a mouse [1].

It is believed that from the early stage throughout life electrical signals generated by the brain represent not only the brain function but also the status of the whole body. This assumption provides the motivation to detect the electric signals that are produced by neurons (which are in the range of μV) by placing electrodes on the surface of scalp and record the brain signals and analyze the signals using

advanced digital signal processing methods to preprocess the raw data measured over the brain as a human subject.

Signal acquisition is done by two methods i.e. invasive and non-invasive. The electrodes are placed inside the brain in invasive and the electrodes are placed on the scalp of the brain in non-invasive. EEG provides a direct measure of cortical activity with millisecond temporal resolution using non-invasive recording. There are two methods of the EEG recording: unipolar and bi-polar. The human brain has two hemispheres left and right. Each Hemisphere has been divided into small lobes and it has its own function and activation pattern for the particular mental activities. The primary motor cortex area of the brain over the central lobe of each hemisphere is responsible for the event related desynchronization (ERD) [2] and event related synchronization (ERS). These patterns ERD and ERS are recorded by placing the sensors over the primary motor cortex area at channels C3, C4 and Cz defined by 10-20 electrode placement system [3].

There are two approaches to analyze the incoming EEG signals from the brain such as online analysis (sequential processing or real time) and offline analysis (batch processing). An online is a real time processing of EEG signals to extract the hidden brain pattern to control an external device. An offline analysis is done for recorded EEG signals to extract hidden information and validate the performance of existing signal processing algorithm as well as the performance of proposed novel algorithm. Different types of classification algorithms were surveyed and it was found that many numbers of algorithms can be applied to the BCI system. Classifiers are placed into five different categories such as linear classifier, [4] nonlinear Bayesian classifier, nearest neighbor classifier, the combination of these classifiers, neural networks. We didn't use neural networks because its assembly of several artificial Neurons which enables to produce nonlinear decision boundaries. LC has very low computational requirement which makes it suitable for the online BCI system. Moreover LC is simple to use and generally provides good results [5].

LC has two types one is LDA (Linear Discriminant Analysis) and another is SVM (Support Vector Machine). Here in this work we are considering LDA classification method based on linear least square methods and Fisher criterion. In this paper, the author considers Fisher criterion based linear discriminant analysis which is most commonly known as Fisher linear discriminant analysis (FLDA) method [6].

II. DATASET DISCRIPTION

In this proposed work, the BCI competition-II Graze dataset was considered. These datasets consist of single normal subject (female, 25 years). The EEG signals were recorded to control the feedback bar by the movement imagination based on the random cues indicating direction of movement left and right on the computer screen [7]. In this dataset, the channels C3, C4 and Cz are derived bipolar by placing sensors 2.5cm anterior and 2.5cm posterior to the position C3, C4 and Cz defined by 10-20 electrode placement system respectively. The sensor layout is shown in fig.1

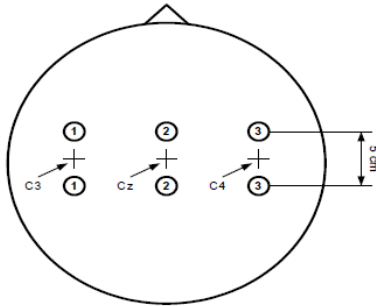


Fig.1: Sensor layout

The data was recorded for 9s and the sampling frequency was 128Hz and it was filtered between 0.5 to 30Hz. In C3, C4 and Cz total 280 trails were recorded in each. The total 140 trails were used for training and another half 140 trails used for testing. The total number of sample points were 1152. During zero to three seconds normal activity of the brain were recorded. At 3s the random cue is indicating direction either left or right appeared on the computer screen. The subject was asked to think to move her left and right hand based on indicated direction. So the useful information to discriminate left and right hand movement imagery lies in the range of 3s to 9s which is shown in fig.2. Thus the author concentrates on 6s data out of total 9s recorded EEG signals to discriminate left and right hand movement imagery [7].

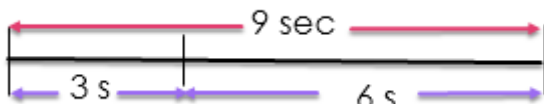


Fig.2: Time division of input data

The recorded signal for left hand movement imagination over channel C3 and C4 are shown in fig.3

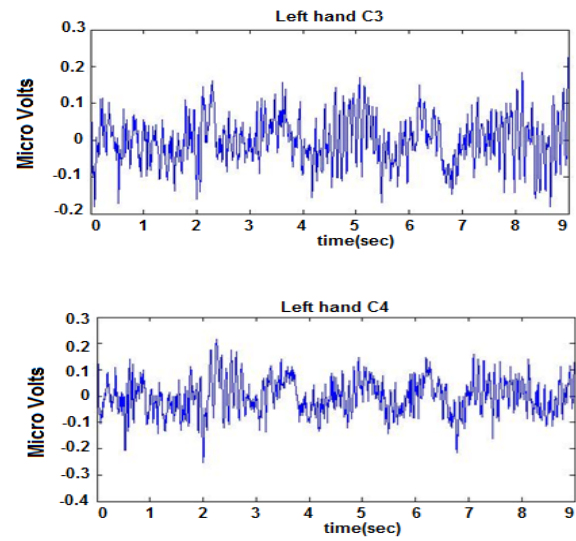


Fig.3: Raw training data 1st trail

III. PROPOSED METHODOLOGY

As we have mentioned there are two approaches mainly used for EEG based BCI system to analyze incoming brain signal. In this paper, the author concentrates on batch processing using unsupervised learning to classify mental thought.

The block diagram of proposed method is shown in fig.4

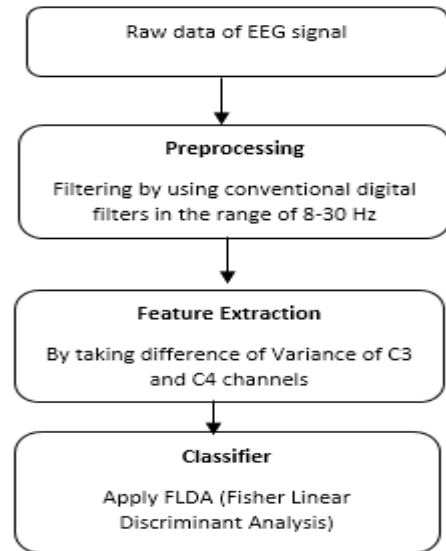


Fig. 4: Proposed method for signal classification

In the proposed method, the raw EEG signals were taken and filtered by mainly two types of filters finite impulse response (FIR) and infinite impulse response (IIR). In the IIR filter, the chebyshev and elliptic filter have been considered.

The filtered EEG signals were segmented to 625ms because authors assume that the EEG signals are non-stationary [8]. The segmented EEG signals over the channels C3 and C4 were selected for calculating the statistical parameter variance as a feature value. All trails were passed through the same procedure and estimated a feature value. The estimated feature value have been concatenated to form a feature matrix for training data. The same approach has been applied to estimate a feature matrix of testing data. After filtering the raw EEG signals were segmented[9]. The segment of 625ms i.e. 80 samples were taken to estimate variance as a feature value in the time domain. The variance is a statistical parameter which captures the deviation from the mean value of a random data. The variance of the random signal X can be expressed as

$$\sigma^2 = Var(X) = E(X^2) - \mu^2 \quad (1)$$

Where, μ is a mean of X. EEG signals are non-stationary random signal so its variance also changes with time. So this fact has been applied to extract unique characteristics from the recorded EEG signals.

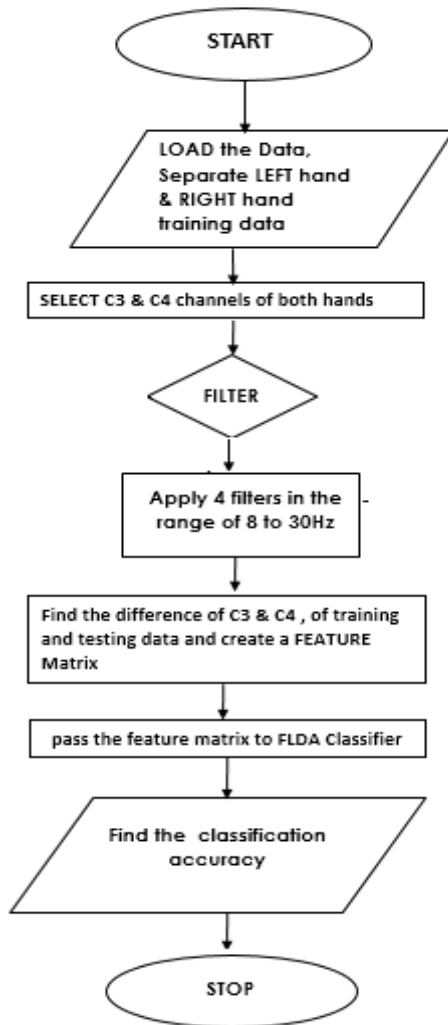


Fig. 5: Flow chart of proposed methodology

Further the obtained feature matrix were passed through the FLDA technique to identify their classes. The feature matrix obtained over the training data were applied to FLDA mechanism to estimate the classifier weight parameter. The estimated classifier weight parameter was applied to classify the testing data feature matrix. The flow chart of proposed steps is shown in fig.5 in order to get the classifier weight parameter and classification accuracy.

As we know that the movement imagery related information contained in the alpha and the beta rhythm of the brain signals. So the frequency range 8-30 Hz has been considered to filter the raw EEG signals. A bandpass filter of pass band 8-30Hz has been considered. The order N of the filter is determined by MATLAB function 'ellipord' with following specifications [10] [11],

$$N = \text{ellipord}(W_p, W_s, R_p, R_s) \quad (2)$$

where, $W_p \rightarrow$ pass band corner frequency $\rightarrow [8 \ 30]/(fs/2)$
 $W_s \rightarrow$ stop band corner frequency $\rightarrow [7.5 \ 30.5]/(fs/2)$
 $R_p \rightarrow$ Pass band ripples $\rightarrow 0.1\text{dB}$
 $R_s \rightarrow$ stop band attenuation $\rightarrow 30\text{dB}$
 $fs \rightarrow$ sampling Frequency $\rightarrow 128\text{Hz}$

Based on these specifications, the estimated filter order is 6. The same filter order is applied for all proposed conventional filters. The variance of C3 and C4 of left and right hand data are plotted in fig.6.

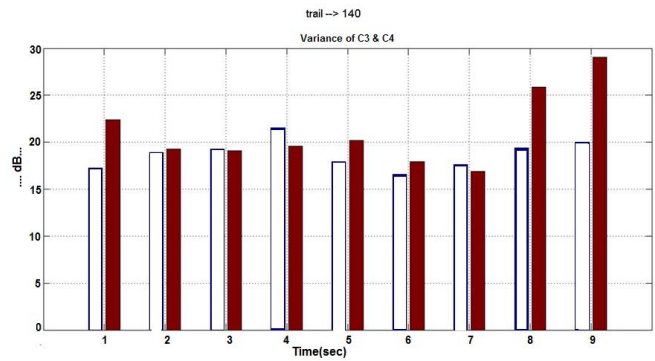


Fig.6: Plot of variance of C3 and C4 for Testing data

IV. RESULTS AND DISCUSSION

The proposed method has been evaluated by the BCI performance parameter classification accuracy. The four filters were applied to filter the raw EEG signals. All the analysis has been done using MATLAB and Biosig toolbox [12]. The performance of the proposed method is listed in the table.1 as a percentage of classification accuracy. The same result has been presented as a bar graph in fig.7.

Table.1: CHECKING CLASSIFICATION ACCURACY WITH DIFFERENT TYPE OF FILTERS

Filter type	Training Classification Accuracy	Testing Classification Accuracy
Chebyshev filter	81.4%	84.29%
FIR bandpass filter	82.9%	88.57%
Butterworth filter	83.6%	86.43%
Elliptic filter	84.3%	86.45 %

The Elliptic filter achieves 84.3% classification accuracy for training data and 86.45% for testing data between the left and right hand movement imagination.

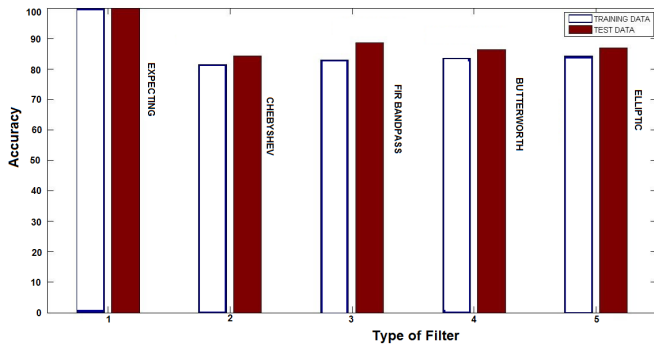


Fig.7: Comparison between different filters with 100%

V. CONCLUSIONS AND FUTURE WORKS

In this paper work, the author finds the effectiveness of conventional filter over classification accuracy. All these conventional filters are of classical filters. Still the proposed filter can't achieve 100% classification accuracy under unsupervised condition. In future, the author will address a new filtering or processing algorithm which will enhance classification accuracy and other performance parameters by developing a spatial filtering and probabilistic filter using the concept of quantum mechanics [13] to achieve 100% classification accuracy under the unsupervised learning. This paper work considers single subject data for analysis. In future

multiple subject can be considered to check the effectiveness of the proposed algorithm.

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